

Data Transfer

In Serial Communication, the word of 8 bits is sent sequentially and received only after each bit has been sent, one at a time and the bits are then assembled back as a single byte just like it was sent.

In parallel communication, all 8 bits are sent simultaneously through 8 different channels where every channel transmits a bit of data, and a byte is received simultaneously.

Speed

Reading the above heading, one can easily contemplate that parallel transmission is faster than its serial counterpart because a serial communication device sends the data in bits and then at the end harmonises the data to form a byte while in parallel communication the same amount of data is sent and received simultaneously.

Connection

Serial transmission uses fewer wires for transmission than Parallel transmission, and this usage of lesser wires makes the signals clearer as compared to parallel communication which is why serial communication is preferred over parallel communication.

Cost and Space

Serial Communication is connected to a single wire or port while a parallel port is usually connected to multiple wires or ports. A serial communication, therefore, requires less space than parallel communication based on the investment done in connecting wires.

Therefore, parallel communication requires more space to accommodate all the wires while serial communication requires minimal space just to connect a single wire to a single connector.

Basic Circuits

Now that we have understood the Basics of an Electronic Circuit and also looked at the two modes of communication in PCB circuits, we'll now look into some of the basic circuits.

1. Rain Alarm

The Rain alarm is used to give an alert about when it is going to rain. The circuit is used in households to guard the washed clothes and other vulner-